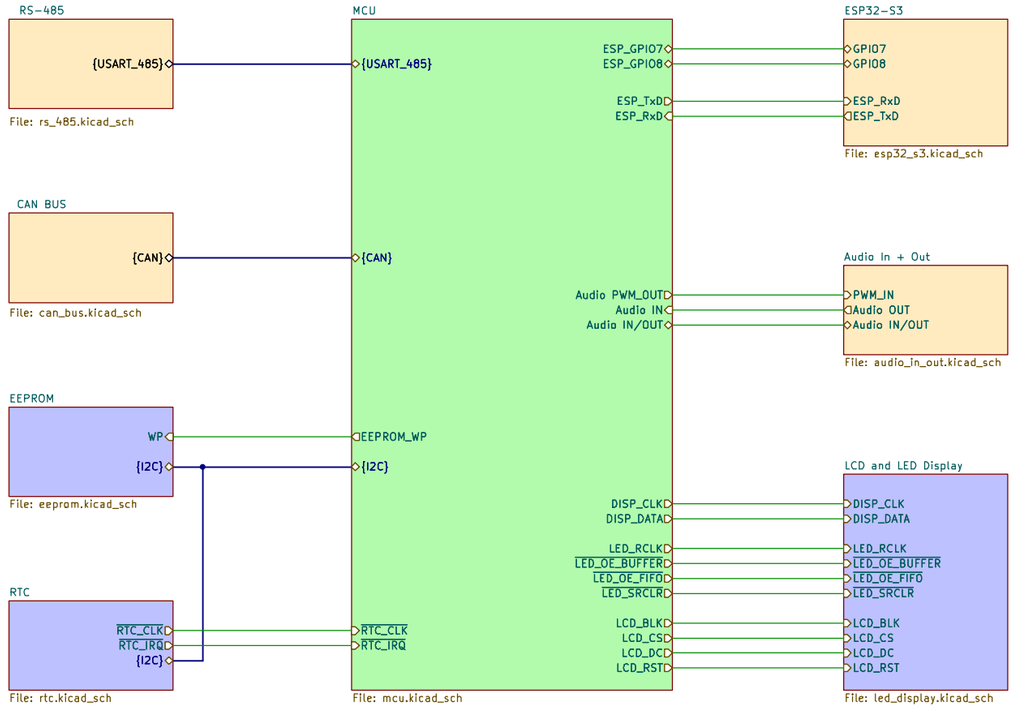
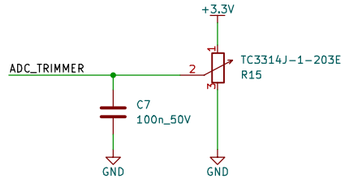
```
AMP SHIELD 100A - PIN MAPPING (STM32F446RE / STM32F401RE, shared layout)
================================================================================
Pin    Signal          F446        F401        Note
--------------------------------------------------------------------------------
Buttons
PA0    BTN1            TIM2_CH1    TIM2_CH1    Input capture
PA1    BTN2            TIM2_CH2    TIM2_CH2    Input capture
PB10   BTN3            TIM2_CH3    TIM2_CH3    Input capture
--------------------------------------------------------------------------------
Status LEDs
PC0    LED1            GPIO_Out    GPIO_Out
PC1    LED2            GPIO_Out    GPIO_Out
PC2    LED3            GPIO_Out    GPIO_Out
PC3    LED4            GPIO_Out    GPIO_Out
--------------------------------------------------------------------------------
LED display (7-seg / 595)
PC10   DISP_CLK        SPI3_SCK    SPI3_SCK    Shared bus with LCD
PC12   DISP_DATA       SPI3_MOSI   SPI3_MOSI   Shared bus with LCD
PA15   LED_RCLK        SPI3_NSS    SPI3_NSS    HW NSS, latch = SPI disable after frame
PC8    LED_SRCLR       GPIO_Out    GPIO_Out    595 clear (active low)
PC9    LED_OE_FIFO     GPIO_Out    GPIO_Out    595 output enable
PB1    LED_OE_BUFFER   TIM3_CH4    TIM3_CH4    Brightness PWM (LVC8T245 OE)
--------------------------------------------------------------------------------
LCD display
PC10   DISP_CLK        SPI3_SCK    SPI3_SCK    Shared bus with LED display
PC12   DISP_DATA       SPI3_MOSI   SPI3_MOSI   Shared bus with LED display
PC11   LCD_CS          GPIO_Out    GPIO_Out    Chip select (SW)
PD2    LCD_DC          GPIO_Out    GPIO_Out    Data/command
PB4    LCD_RST         GPIO_Out    GPIO_Out    Reset
PB0    LCD_BLK         TIM3_CH3    TIM3_CH3    Backlight PWM
--------------------------------------------------------------------------------
Audio
PA4    AUDIO_IN/OUT    DAC_OUT1    ADC1_IN4    In/Out (F446) / In (F401), jack
PA6    AUDIO_IN        ADC1_IN6    ADC1_IN6    Audio input
PA8    AUDIO_PWM_OUT   TIM1_CH1    TIM1_CH1    PWM -> LPF -> LM386 (primary on F401)
PA11   BUZZER_PWM      TIM1_CH4    TIM1_CH4    Buzzer via Q1, jumper P3
--------------------------------------------------------------------------------
I2S socket (P2)
PB12   I2S_LRCL        I2S2_WS     I2S2_WS
PB13   I2S_BCLK        I2S2_CK     I2S2_CK
PB15   I2S_SD          I2S2_SD     I2S2_SD
PB2    I2S_SEL         GPIO_Out    GPIO_Out    Socket select
--------------------------------------------------------------------------------
WS2812
PA7    WS_DIN          SPI1_MOSI   SPI1_MOSI   TX-only + DMA
                       TIM14_CH1   TIM3_CH2    Alternate to SPI using TIM3/14
--------------------------------------------------------------------------------
RS-485
PA9    MCU.TxD         USART1_TX   USART1_TX   ST3485
PA10   MCU.RxD         USART1_RX   USART1_RX   ST3485
PA12   MCU.DE          GPIO_Out    GPIO_Out    DE + /RE (SW timing)
--------------------------------------------------------------------------------
CAN (F446 only)
PB6    CAN_TX          CAN2_TX     (unused)
PB5    CAN_RX          CAN2_RX     (unused)
--------------------------------------------------------------------------------
I2C (EEPROM + RTC)
PB8    I2C.SCL         I2C1_SCL    I2C1_SCL
PB9    I2C.SDA         I2C1_SDA    I2C1_SDA
PB14   EEPROM_WP       GPIO_Out    GPIO_Out    Write protect
--------------------------------------------------------------------------------
ESP32-S3 (XIAO)
PC6    ESP_TxD         USART6_TX   USART6_TX
PC7    ESP_RxD         USART6_RX   USART6_RX
PC5    ESP_GPIO8       GPIO        GPIO
PH1    ESP_GPIO7       GPIO        GPIO        Free in HSE bypass mode
--------------------------------------------------------------------------------
Measurement
PC4    ADC_TRIMMER     ADC1_IN14   ADC1_IN14   Trimmer
PB7    F_MEASURE       TIM4_CH2    TIM4_CH2    Input capture, jumper P4
--------------------------------------------------------------------------------
Debug / VCP
PA13   TMS             SWD         SWD
PA14   TCK             SWD         SWD
PB3    SWO             SWO         SWO         Trace
PA2    VCP_TX          USART2_TX   USART2_TX   ST-LINK console
PA3    VCP_RX          USART2_RX   USART2_RX   ST-LINK console
--------------------------------------------------------------------------------
Clocks / System
PH0    MCO_8MHz        OSC_IN      OSC_IN      HSE bypass, ST-LINK MCO (SB closed)
PC13   RTC_IRQ         EXTI13      EXTI13      External IRQ from RX8111
PC14   RTC_CLK         OSC32_IN    OSC32_IN    LSE bypass, RX8111 CLKOUT via R13
PC15   -               (free)      (free)      Analog/NC
PA5    -               GPIO_Out    GPIO_Out    Nucleo LD2, not routed to shield
================================================================================
```



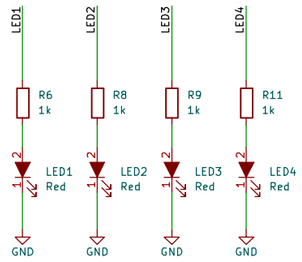
J1 FiducialMark
J2 FiducialMark
J3 FiducialMark

CTU in Prague
Sheet: /
File: amp_shield.kicad_sch
Title: AMP Development Kit
Size: A3 | Date: 2026-07-09 | Rev: 100A
KiCad E.D.A. 10.0.3 | Id: 1/9

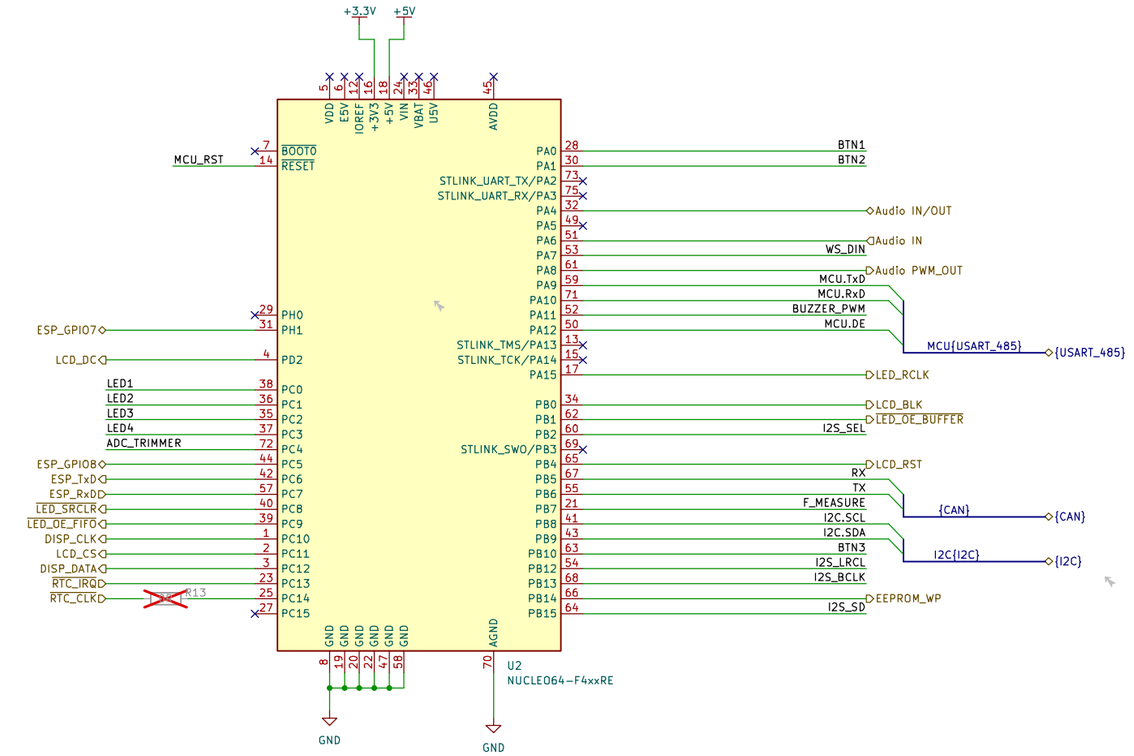
Trimmer for ADC



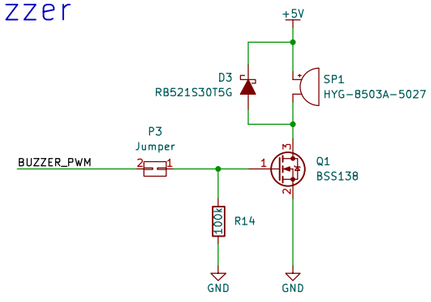
LEDs



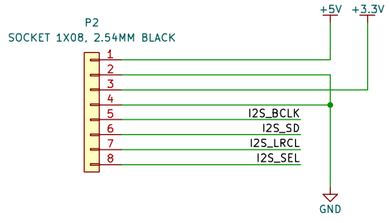
Microcontroller – Nucleo F4xxRE



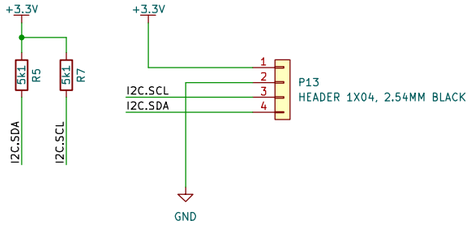
Buzzer



I2S Header for Microphone



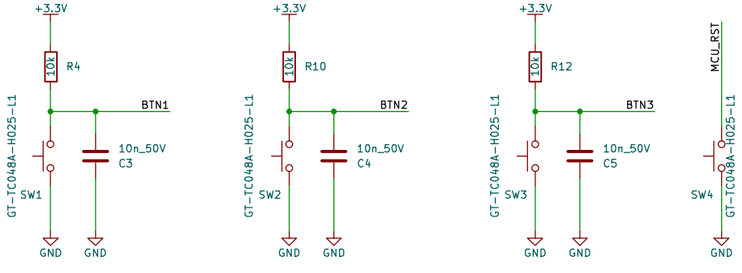
I2C Header



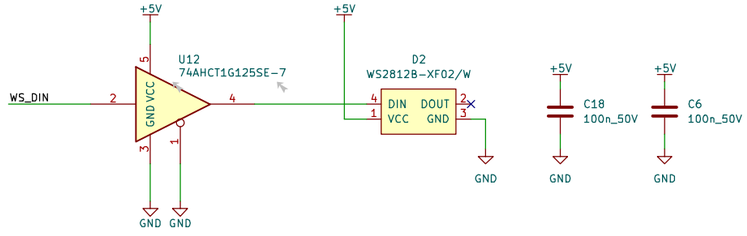
Header for Frequency Measurement



Buttons



LED WS2812 with Level Shifter



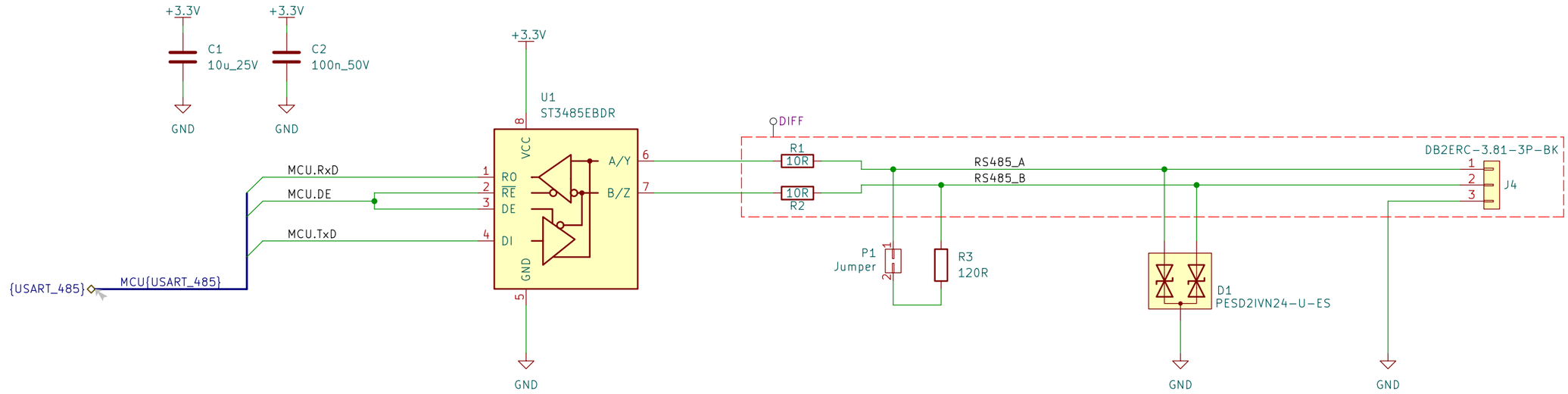
CTU in Prague
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File: mcu.kicad_sch

Title: AMP Development Kit – Microcontroller

Size: A3	Date: 2026-07-09	Rev: 100A
KiCad E.D.A. 10.0.3		Id: 2/9

Q2A
BSS138PS
Q2B
BSS138PS
Q3A
BSS138PS
Q3B
BSS138PS
GND
a
b
c
d
e
f
g
h (DP)
CC1
CC2
CC3
CC4
SR420361N
U3
R18 220R
R19 220R
R20 220R
R21 220R
R22 220R
R23 220R
R24 220R
R25 220R
U4
SN74LVC8T245PWR
GND
VCCB
VCCB
VCCA
OE
DIR
A8
A7
A6
A5
A4
A3
A2
A1
B8
B7
B6
B5
B4
B3
B2
B1
+3.3V
+5V
C30
1u_25V
C8
100n_50V
C9
100n_50V
C29
1u_25V
LED_OE_BUFFER
R37
10k
U5
74HC595
U6
74HC595
VCC
GND
DS
SHCP
STCP
OE
RST
Q0
Q1
Q2
Q3
Q4
Q5
Q6
Q7
Q7S
C11
100n_50V
C12
100n_50V
R16
10k
R17
10k
C10
1u_25V
DISP_DATA
DISP_CLK
LED_RCLK
LED_OE_FIFO
LED_SRCLR
P5
Display 320x240 socket
GND
VCC
SCL
SDA
RST
DC
CS
BLK
LCD_RST
LCD_DC
LCD_CS
LCD_BLK

CTU in Prague
Sheet: /LCD and LED Display/
File: led_display.kicad_sch
Title: AMP Development Kit – LED/LCD Display
Size: A3
Date: 2026-07-09
Rev: 100A
KiCad E.D.A. 10.0.3
Id: 3/9

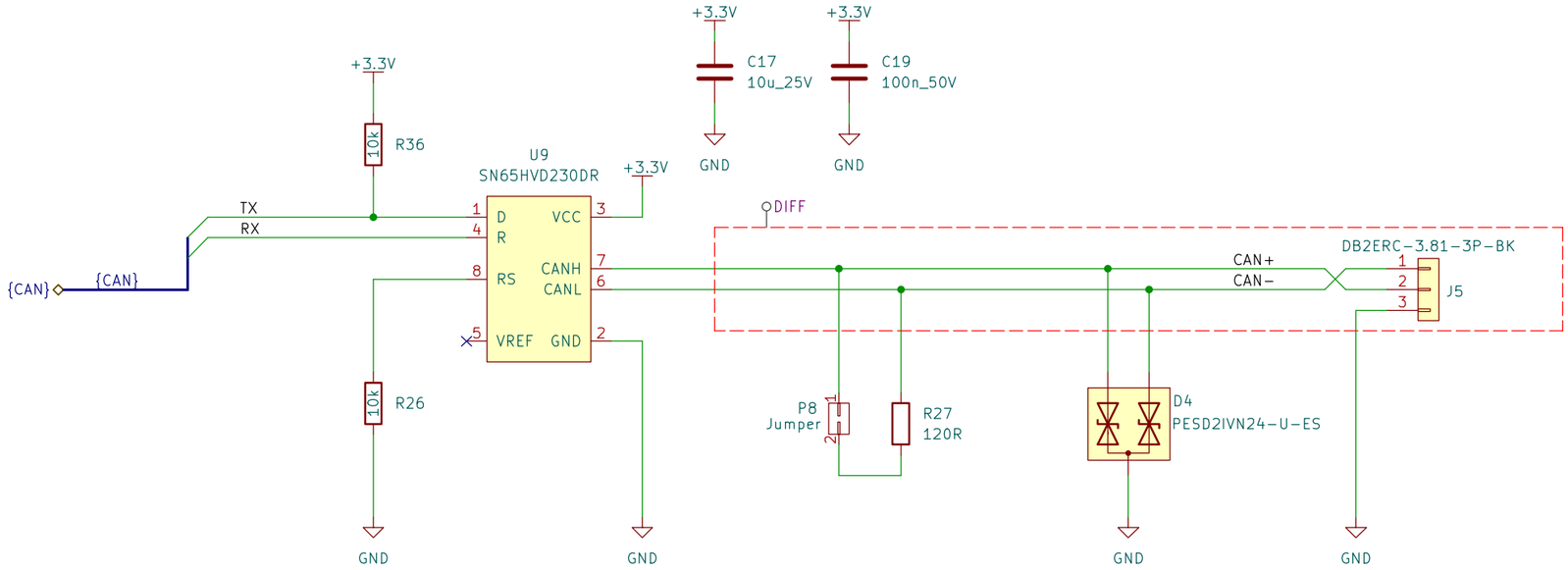


CTU in Prague

Sheet: /RS-485/
File: rs_485.kicad_sch

Title: AMP Development Kit – RS485

Size: A4	Date: 2026-07-09	**Rev: 100A**
KiCad E.D.A. 10.0.3		Id: 4/9

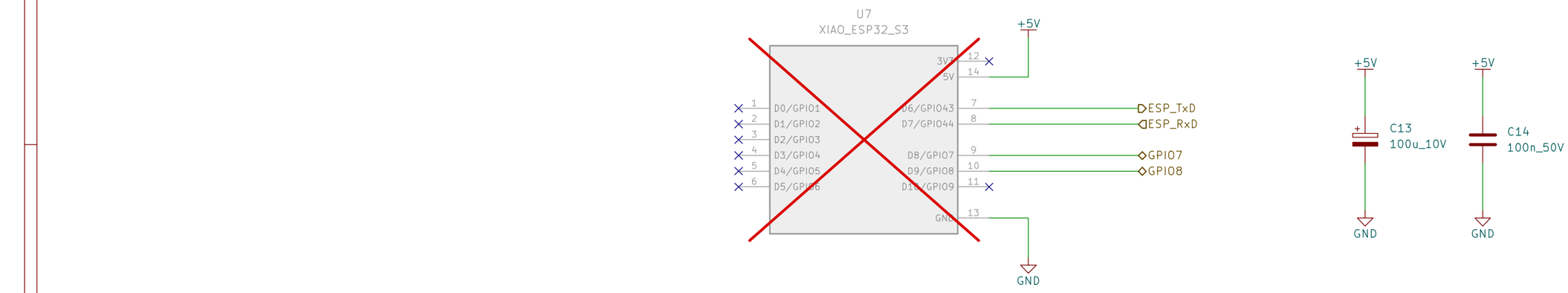


CTU in Prague

Sheet: /CAN BUS/
File: can_bus.kicad_sch

Title: AMP Development Kit – CAN Bus

Size: A4	Date: 2026−07−09	**Rev: 100A**
KiCad E.D.A. 10.0.3		Id: 5/9

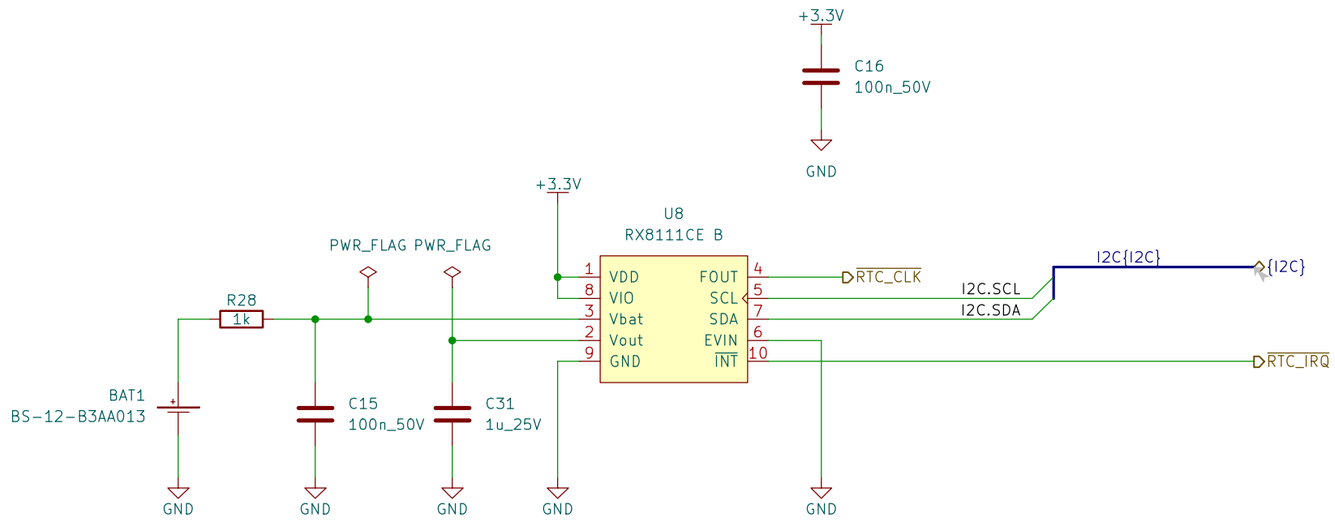


CTU in Prague

Sheet: /ESP32-S3/
File: esp32_s3.kicad_sch

Title: AMP Development Kit – ESP32-S3 Gateway

Size: A4	Date: 2026-07-09	**Rev: 100A**
KiCad E.D.A. 10.0.3		Id: 6/9

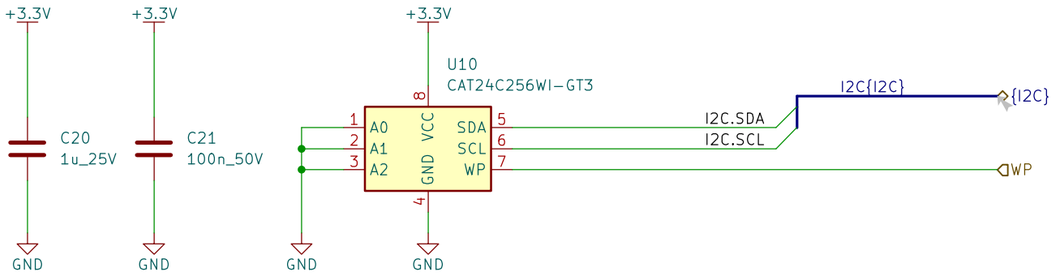


CTU in Prague

Sheet: /RTC/
File: rtc.kicad_sch

Title: AMP Development Kit – RTC

Size: A4	Date: 2026-07-09	**Rev: 100A**
KiCad E.D.A. 10.0.3		Id: 7/9



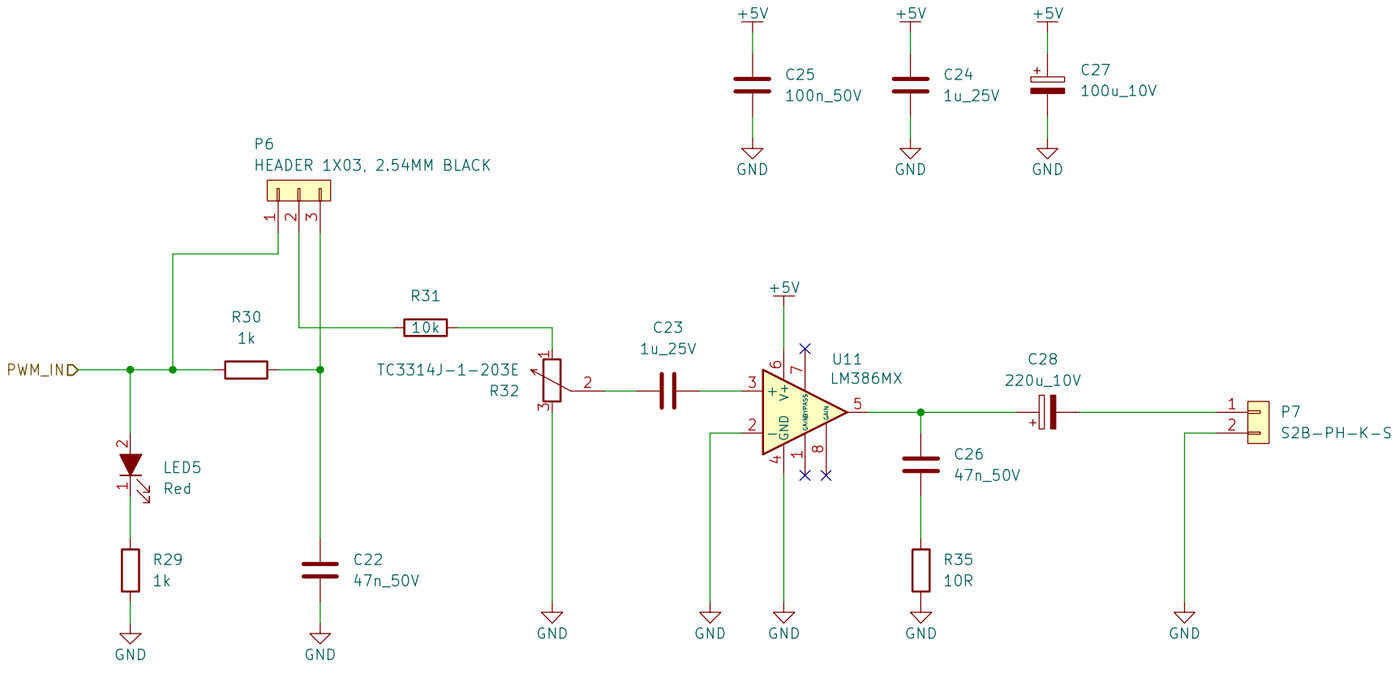
Alternative on LCSC C7432367

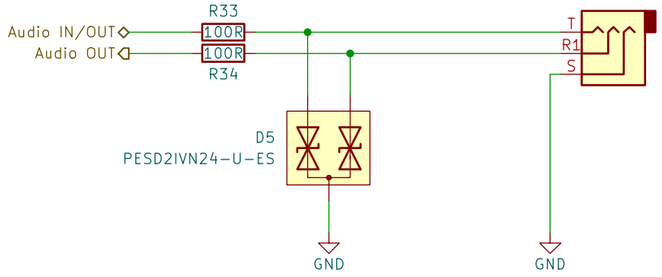
CTU in Prague

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Title: AMP Development Kit – EEPROM

Size: A4	Date: 2026-07-09	**Rev: 100A**
KiCad E.D.A. 10.0.3		Id: 8/9





CTU in Prague

Sheet: /Audio In + Out/
File: audio_in_out.kicad_sch

Title: AMP Development Kit – Audio In + Out

Size: A4	Date: 2026-07-09	**Rev: 100A**
KiCad E.D.A. 10.0.3		Id: 9/9